NYC Airbnb Listings — Reproducible Data Workflow

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Overview

This module implements a complete, reproducible data workflow using the NYC Airbnb Listings dataset from Kaggle. The workflow loads the dataset, applies structured cleaning functions, performs exploratory data analysis, and produces multiple labeled visualizations. The design emphasizes reproducibility, modular function design, and transparent documentation—forming a foundation for later machine learning, deep learning, generative AI, and agentic systems.

Dataset Description

The dataset contains **47,735 Airbnb listings** across New York City with **16 columns**, including price, room type, neighbourhood group, availability, and review metrics. Key variables used in this workflow include:

- **price** — nightly rate
- **neighbourhood_group** — borough-level grouping
- **room_type** — property type
- **minimum_nights** — booking requirement
- **availability_365** — annual availability
- **number_of_reviews** — listing activity indicator

The dataset is large enough for robust descriptive analysis and contains meaningful categorical and numeric variables suitable for grouping, visualization, and statistical exploration.

Workflow Description

The workflow proceeds through five structured stages:

1. Ingestion

The CSV is loaded using Pandas, and the first rows are displayed to confirm correct data ingestion.

2. Cleaning

Two reusable cleaning functions (each with docstrings) handle structural normalization, missing values, and outlier removal.

- Price outliers are clipped at the **0.5% and 99.5% quantiles**.
- Non-rentable listings (shared rooms) are removed.

3. Exploratory Analysis

EDA includes:

- Spearman correlations for numeric → price
- Correlation ratio (η) for categorical → price
- Grouped descriptive statistics by borough and property type

4. Visualizations

Four labeled visualizations summarize univariate and bivariate patterns:

- Figure 5.1 Distribution of Price, Neighbourhood Group, and Room Type
- Figure 5.2.1 Median price by neighbourhood group
- Figure 5.2.2 Price distribution by neighbourhood group (violin plot)
- Figure 5.2.3 Price by property type across neighbourhood group (facet boxplots)

(Note: Figure numbering is kept consistent with **data_workflow.ipynb** for easy reference)

5. Summary

Findings, assumptions, limitations, and surprising patterns are documented.

Key Decisions and Assumptions

Cleaning Decisions

Outlier clipping at the 0.5% and 99.5% quantiles removes extreme luxury listings that distort correlations and boxplots.

Dropping shared rooms narrows the analysis to private rooms and entire homes, aligning the workflow with typical Airbnb pricing comparisons.

These decisions reflect best practices: missing or extreme values can distort statistical summaries and introduce bias if not handled carefully. As Little & Rubin (2002) note, "common imputation or deletion approaches can shift results and amplify bias when data are not missing at random." This risk is explicitly acknowledged in the workflow.

EDA Focus

The EDA block answers a single question:

How much of price does each feature explain on its own?

- **Spearman's ρ** measures monotonic numeric association.
- **Correlation ratio η** measures how well categorical grouping predicts price.
- Squaring each statistic (ρ^2, η^2) places all features on a common scale: **proportion of variance explained**.

Visualization Intent

- **Figure 5.2.1**: Reveal the borough-level price gradient.
- **Figure 5.2.2**: Show distribution shape and skew across boroughs.
- **Figure 5.2.3**: Demonstrate property-type price stratification within each borough.

(**Note:** Figure numbering is kept consistent with **data_workflow.ipynb** for easy reference)

Summary & Interpretation

Variance Explained (Univariate)

Feature	Statistic	Variance Explained
minimum_nights	ρ = 0.098	0.96%
availability_365	ρ = 0.099	0.98%
number_of_reviews	ρ = –0.054	0.30%
neighbourhood_group	η = 0.088	0.77%
room_type	η = 0.219	4.8%

Interpretation: Every feature except **room_type** explains **under 1%** of price variance. Room type is the strongest single predictor (~5%), confirming that entire homes/apt listings consistently outprice private rooms.

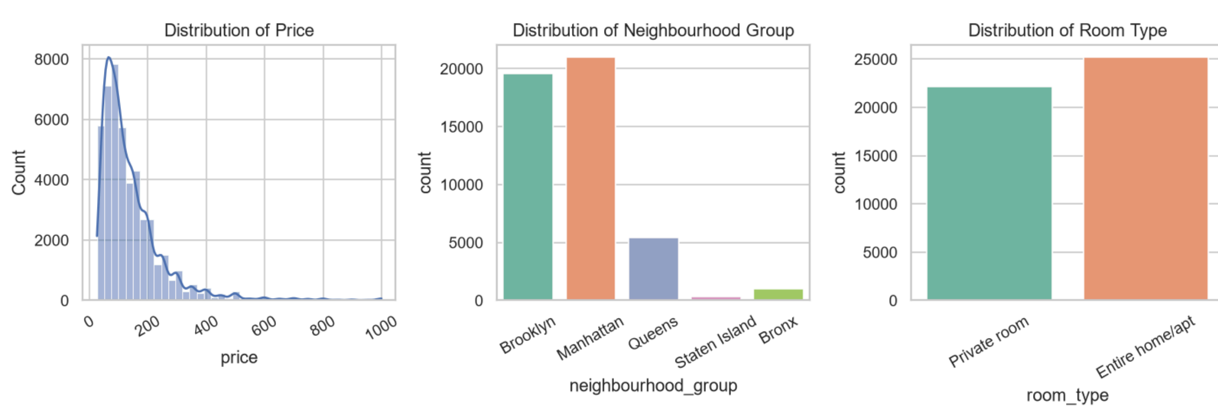

Figure 5.1 Distribution of Price, Neighbourhood Group, and Room Type

Location gradient (Figures 5.2.1–5.2.2):

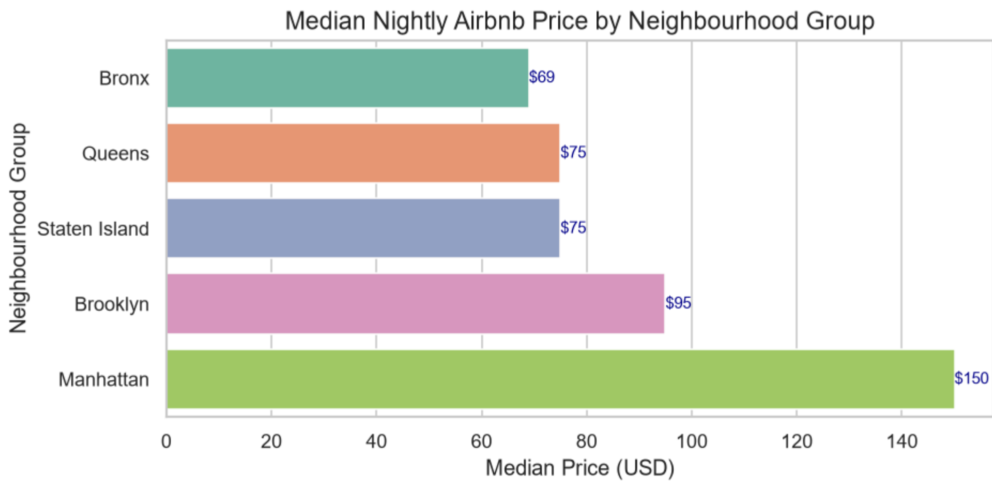

Figure 5.2.1 Median Nightly Price by Neighbourhood Group

- Manhattan: **$150 median**
- Brooklyn: **$95**
- Queens: **$75**
- Staten Island: **$75**
- Bronx: **$69**

Manhattan's mean price (~$181) is nearly double the Bronx's (~$88), reflecting demand concentration and luxury inventory.

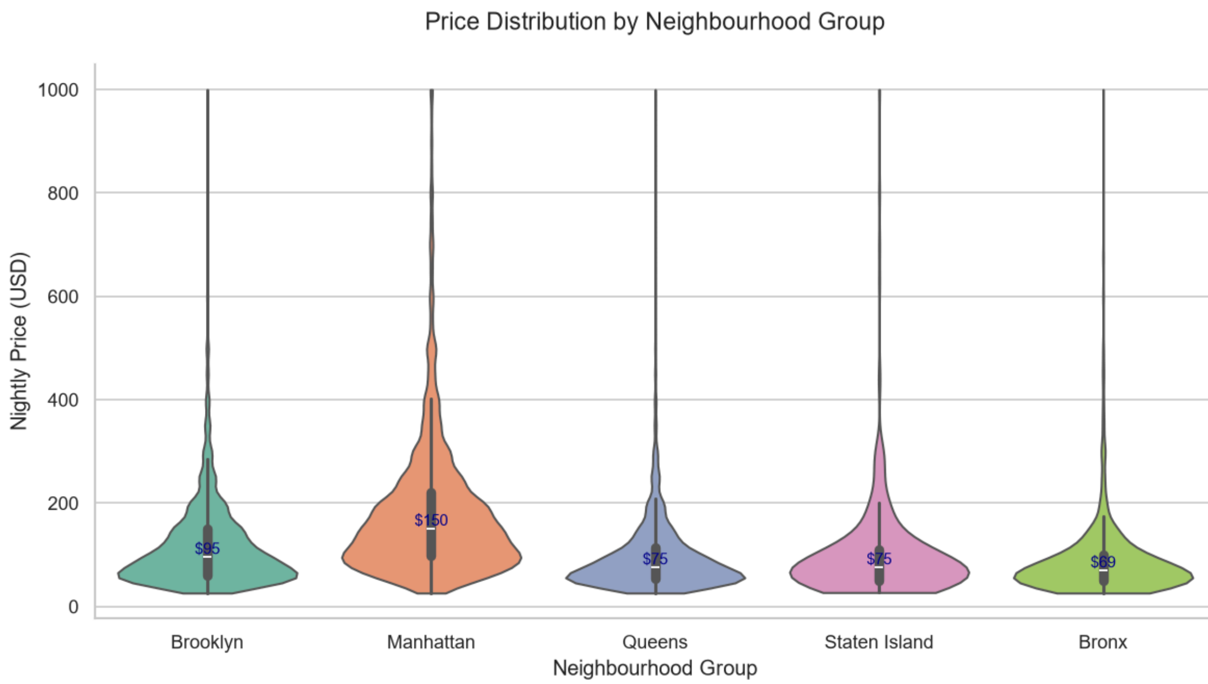

Figure 5.2.2 Price Distribution by Neighbourhood Group

Every borough's price distribution is strongly right-skewed, confirming the long upper tail seen in the histogram. Manhattan and Brooklyn show the widest price distributions (supporting a range of listings from budget to luxury), whereas the Queens, Bronx and Staten Island are tightly clustered around lower prices.

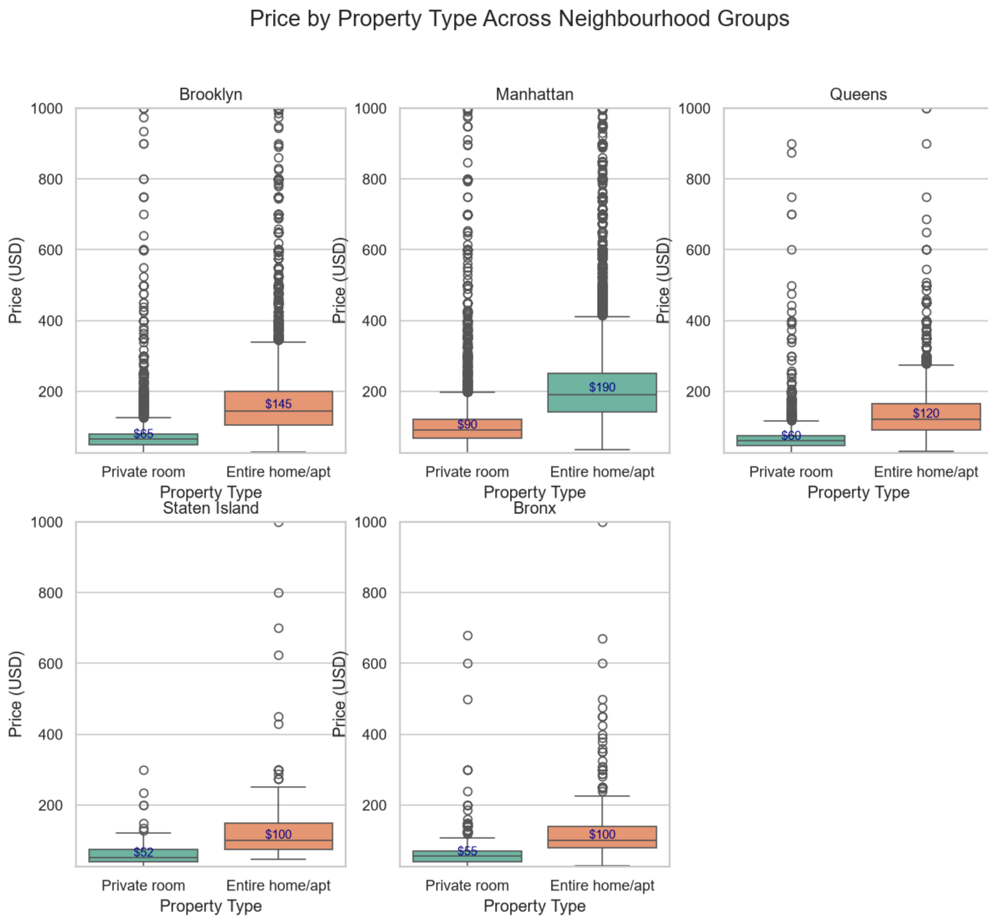

Fig 5.2.3 Price by Room Property Type across Neighbourhood Groups

Property-Type Stratification (Figure 5.2.3)

Across all boroughs:

- **Entire home/apt** listings have significantly higher medians than **private rooms**.
- The gap is largest in Manhattan, where entire-home listings reach luxury price levels.

Why is the neighbourhood η^2 so small?

Despite visible median gaps, η^2 is only **0.77%** because:

- Price is heavily right-skewed.
- Extreme values inflate the denominator of η.
- η is computed on **means**, which are unstable under skew.

As Wilcox (2012) explains, medians are more robust for skewed distributions—hence their use in grouped statistics and plots.

Takeaway

Price is influenced by:

- **Property type** (strongest)
- **Location** (secondary)
- **Other features** (minimal)

Even the strongest feature explains <5% of variance, indicating that price is driven more by **within-group spread** than by any single variable. A predictive model would require additional features beyond these univariate measures.

Responsible Practice (Bias and Data Quality)

Potential sources of bias include:

- **Outlier clipping** removes ~1% of listings, excluding luxury properties from summary statistics.
- **Dropping shared rooms** narrows the population to private and entire-home listings, limiting generalizability.
- **Small sample sizes** in Bronx and Staten Island make medians less stable.
- **Missing reviews** may reflect listing age or inactivity, not quality.

These risks are documented, and future workflows could incorporate sensitivity analysis or alternative imputation strategies.

Reproducibility

The workflow is fully reproducible:

- **requirements.txt** was generated using `pip freeze`, pinning all dependencies.

- The GitHub repository includes multiple commits and a dedicated feature branch, supporting auditability and version control.
- Modular cleaning and EDA functions ensure reusability and transparency.

As Danchev (2022) notes, reproducible workflows require explicit environment capture and modular code design—principles followed throughout this project.

Conclusion

This module establishes a clean, reproducible data workflow that meets all rubric requirements. It provides a strong foundation for future modules involving machine learning, deep learning, generative AI, and agentic systems by ensuring the dataset is well-understood, well-cleaned, and well-documented.

References

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